

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 2: Univariate Time Series (1)

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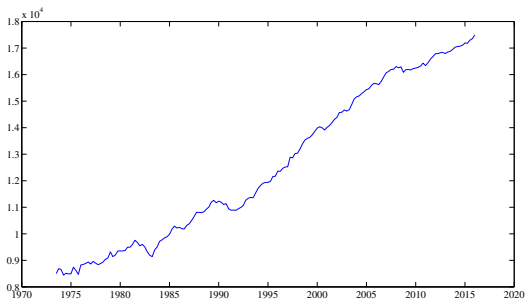
March 4, 2019

Univariate Time Series Models

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	1, 2	HB139 .E55 2015
Verbeek	A Guide to Modern Econometrics	2, 3, 4, 5	HB139 .V465 2012

Australian real GDP per capita (seasonally adjusted)



A Time Series Process

Definition.

A **stochastic process** is a collection of random variables that are ordered in time.

A **time series** is a **stochastic process**

- each observation is a **random variable**
- observations evolve in time according to some probabilities

Realizations, Moments

Definition.

A **realization** is one of a (typically) *infinite* set of values of $y_t, t = 1, \dots, T$, randomly generated according to the probability distribution underlying the stochastic process.

We are typically interested in the following moments characterizing the probability distribution:

- **Mean:** $\mu_t = E(y_t), t = 1, \dots, T$; which can be interpreted as the average value of y_t taken over all possible realizations.
- **Variance:** $\text{Var}(y_t) = E((y_t - \mu_t)^2), t = 1, \dots, T$; i.e., the average of squared deviations from the mean.
- **Covariance:** $\text{Cov}(y_t, y_{t-k}) = E((y_t - \mu_t)(y_{t-k} - \mu_{t-k})), t = k + 1, \dots, T$.

Modelling Time-Series Data

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We shall focus on time series models as forecasting tools (expand later).

Modelling Time Series Data

One simple way to model the time series is with the “regression”:

$$y_t = a_0 + a_1 y_{t-1} + u_t.$$

This is called the **first order auto-regressive model**, or AR(1).

We might assume that u_t is an error term satisfying standard CR assumptions.

Given the model specification, we must have random/stochastic regressor.

Estimating the AR(1) Model

What is the implication of having a “stochastic” regressor?

Suppose, for the moment, we knew a_0, a_1 and wanted to forecast y_{T+h} .

A reasonable forecast $\hat{y}_{T+h}$ could be:

$$\begin{aligned} E(y_{T+h} \mid y_1, \dots, y_T) &= E(a_0 + a_1 y_{T+h-1} + u_{T+h} \mid \cdot) \\ &= a_0 + a_1 E(y_{T+h-1} \mid \cdot) + E(u_{T+h} \mid \cdot) \\ &= a_0 + a_1 (a_0 + a_1 E(y_{T+h-2} \mid \cdot) + E(u_{T+h-1} \mid \cdot)) \\ &\vdots \\ &= a_0 (1 + a_1 + a_1^2 + \dots + a_1^{h-1}) + a_1^h y_T \end{aligned}$$

Estimating the AR(1) Model

In the limiting case as $h \rightarrow \infty$, we get:

- $\hat{y}_{T+h} \rightarrow \frac{a_0}{1-a_1}$ if $|a_1| < 1$;
- $\hat{y}_{T+h} \rightarrow \text{indeterminate form}$ (i.e. does not exist) if $|a_1| \geq 1$.

The AR(1) model with $|a_1| \geq 1$ is called **non-stationary**.

- can still proceed with OLS estimation, but non-stationarity causes problems in deriving asymptotic properties

All properties of the AR(1) can be generalized to the AR(p):

$$y_t = a_0 + a_1 y_{t-1} + \cdots + a_p y_{t-p} + u_t.$$

Stationarity

There are several forms of **stationarity** that can be used to describe a stochastic process. We will keep things simple with the following definition.

Definition.

A stochastic process is **stationary** if the data generating process is such that the mean, variance, and covariances are independent of time (time invariant), i.e.

$$\begin{aligned}E(y_t) &= \mu, \\ \text{Var}(y_t) &= \sigma_y^2 = \gamma_0, \\ \text{Cov}(y_t, y_{t-k}) &= \gamma_k, \quad k \geq 1,\end{aligned}$$

hold for all t .

For now, we will only consider stationary processes, and we will relax this assumption later in the course.

Moving Average Models

What if the errors $u_1, \dots, u_T$ are also **correlated**?

Correlated errors could be modeled, for example, by

$$u_t = \varepsilon_t + \beta_1 \varepsilon_{t-1},$$

where ε_t is the **uncorrelated** innovation in the process.

The above is called a **first-order moving average** MA(1) process for u_t .

Can be generalized to MA(q):

$$u_t = \varepsilon_t + \beta_1 \varepsilon_{t-1} + \dots + \beta_q \varepsilon_{t-q}.$$

The Autoregressive Moving Average Model

Putting the AR(1) and and MA(1) together, we get:

$$y_t = a_0 + a_1 y_{t-1} + \beta_1 \varepsilon_{t-1} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ are uncorrelated.

This is called the **autoregressive moving average model** ARMA(1, 1).

Things to note:

- $\beta_1 = 0$ implies an AR(1) process for y_t ;
- $a_1 = 0$ implies a MA(1) process for y_t ;
- if $|a_1| < 1$, then there exists an equivalent pure MA process with infinitely many lags;
- if $|\beta_1| < 1$, then there exists an equivalent pure AR process with infinitely many lags.

The ARMA(p, q)

In general, we work with a model containing p autoregressive lags and q moving average lags, i.e.

$$y_t = a_0 + \sum_{j=1}^p a_j y_{t-j} + \varepsilon_t + \sum_{j=1}^q \beta_j \varepsilon_{t-j}.$$

This turns out to be an extremely flexible way to model **any** stationary time series.

Unless $q = 0$, we cannot use OLS to estimate all parameters. Instead, numeric optimisation is required, but fortunately, Stata (and many other packages) have built-in routines to do this!

Autocovariance and Autocorrelation Functions

Most interesting properties of a time-series can be described by its **autocovariance function** or **autocorrelation function** (ACF).

An **autocorrelation** is just the **autocovariance** scaled by the process **variance**, i.e. $\rho_k = \frac{\gamma_k}{\gamma_0}$.

- The scaling eliminates dependence on the unit of measurement (e.g. γ_k is higher for a process measured in dollars than the same process measures in cents, but ρ_k is the same).
- In general, $\rho_k \in (-1, 1)$.

The ACF is the plot of ρ_k against $k = 1, 2, \dots$. For a **stationary** process, $\rho_k = \rho_{-k}$, and therefore, the ACF describes all the autocorrelations in the process.

Given data that are a realization of a stochastic process, we can estimate the underlying ACF with **sample autocorrelations** to compute the **sample autocorrelation function** (SACF) or **correlogram**.

Partial Autocorrelation Function (PACF)

Another way to measure the relationship between y_t and y_{t+k} is to compute the correlation with the influence of all intermediary variables $y_{t+1}, \dots, y_{t+k-1}$ “filtered out.”

This is unknown as the **partial autocorrelation** ϕ_{kk} ; plotting ϕ_{kk} against k generates the **partial autocorrelation function** (PACF).

Things to note:

- $\phi_{11} = \gamma_1$;
- for an $\text{AR}(p)$, $\phi_{pp} = a_p$ and $\phi_{kk} = 0$ for all $k > p$;
- in general, the PACF can be recovered from the ACF using **Yule-Walker** equations.

Partial Autocorrelation Function (PACF)

Using the Yule-Walker equations, we can work out the PACF from the ACF,

$$\phi_{11} = \rho_1$$

$$\phi_{22} = (\rho_2 - \rho_1^2)/(1 - \rho_1^2)$$

$$\vdots$$

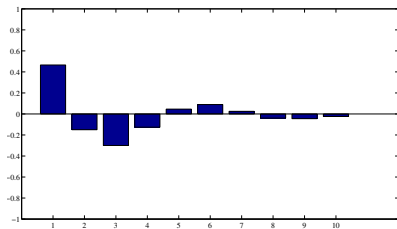
$$\phi_{kk} = \frac{\rho_k - \sum_{j=1}^{k-1} \phi_{k-1,j} \rho_{k-j}}{1 - \sum_{j=1}^{k-1} \phi_{k-1,j} \rho_j}, k = 3, 4, 5, \dots$$

where, $\phi_{k,j} = \phi_{k-1,j} - \phi_{kk} \phi_{k-1,k-j}, j = 1, 2, 3, \dots, k-1$.

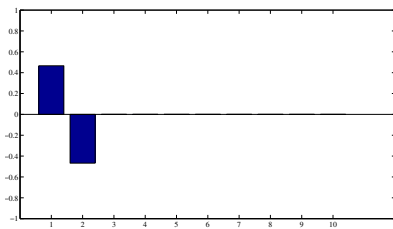
The ACF and PACF of an AR(2) Process

DGP: $y_t = 0.7y_{t-1} - 0.49y_{t-2} + \varepsilon_t$

ACF



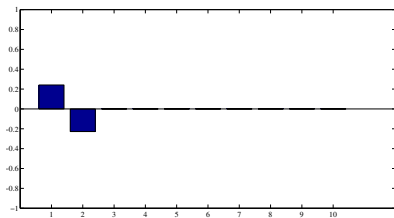
PACF



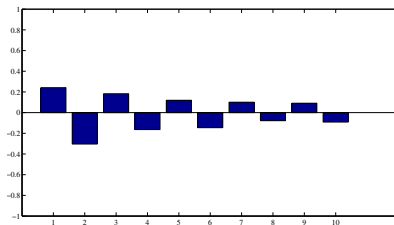
The ACF and PACF of a MA(2) Process

DGP: $y_t = \varepsilon_t + 0.7\varepsilon_{t-1} - 0.49\varepsilon_{t-2}$

ACF



PACF



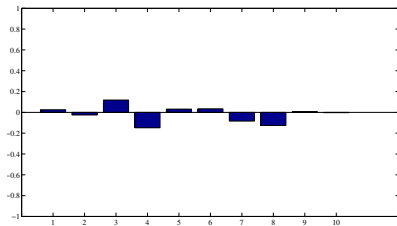
The Sample ACF and PACF for AUS GDP per capita

Let's transform the data by taking **natural logs** and **first differences** (Why?). Then, compute the sample ACF and PACF.

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